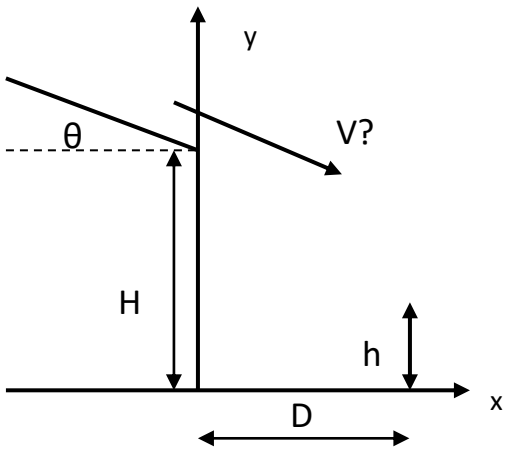
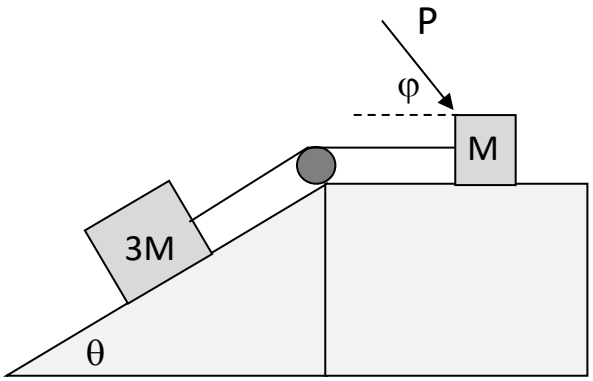
Physics 1135 Test 1 Preparation Homework

1. Students roll a snowball from a roof that makes an angle θ with respect to the horizontal. The snowball leaves the roof with speed *V*, at height *H* above the ground. A physics professor is standing a horizontal distance *D* away from the building; the professor's head is at height *h* above the ground.



a) Complete the diagram with all information necessary for the task below.
b) Derive an expression for the initial speed *V* the snowball must have in order to hit the professor's head.
c) Derive an expression for the speed with which the snowball hits the professor's head. You may use V as a system parameter for this part.

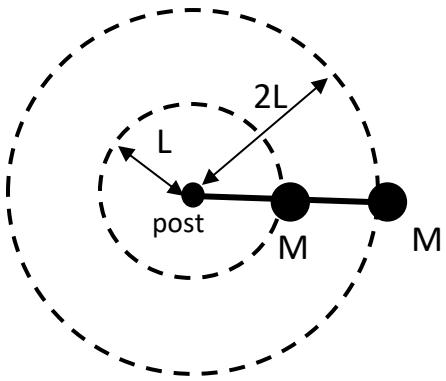
2. A block of mass 3*M* is sliding down a **frictionless incline** that makes an angle θ with the horizontal. It is connected to a second block of mass *M* that is on a **rough horizontal** surface by a massless rope that passes over a frictionless massless pulley. An external force *P* retards the motion by pushing the block at an angle φ with horizontal as shown.



a) On the figure, superimpose fully labeled free-body diagrams for each block, including all information that you need to solve the task below.

b) Derive an expression for the coefficient of friction μ between the horizontal surface and the block of mass *M* that is required for the blocks to move at a constant speed.

3. Two blocks, each of mass *M*, are on a frictionless **horizontal** surface. They are connected to each other by a cord of length *L*. The inner mass is connected to a post by another cord, also of length *L*. The masses rotate about the post in a horizontal circle. The figure shows a view from above. Both masses complete a full rotation around the post in the same time. The speed of the inner block is *V*.



a) Draw two fully-labeled free body diagrams, one for each mass and include all information needed to solve part b).

b) Derive expressions for the tension in each cord, in terms of relevant system parameters.